Dual-Prototype Learning With Multisemantic Fusion for Cross-Domain Few-Shot Hyperspectral Image Classification

Yuhang Li, Jinrong He, Hanchi Liu, and Zhaokui Li, *Member, IEEE*

***Abstract*—Multimodal learning has demonstrated outstanding performance in various fields, including visual tasks, due to its ability to integrate information from different data sources. In the field of hyperspectral image (HSI) classification, although existing research can handle complex image data, the utilization of semantic information is insufficient. Over-reliance on a single image modality overlooks the synergistic effects of cross-modal information, thereby limiting the model's performance in category center representation and discrimination, especially in scenarios with scarce samples. To address these challenges, this article proposes a dual-prototype learning with multisemantic fusion (DPL-MSF) method. Specifically, within the DPL-MSF framework, precise text prototypes are constructed by leveraging category textual information and prompt learning strategies to capture the first layer of semantic information. In addition, soft labels (SLs) are introduced into the image prototypes, and a feature fusion module is designed to deeply integrate SL information, thereby obtaining another layer of semantic information. The primary role of the text prototype is to assist in the generation of image prototypes before the introduction of SLs, through contrastive learning methods. Under the synergistic effect of multiple semantic information, the final class-level prototypes generated are more representative and discriminative, significantly enhancing the performance of few-shot learning. Experiments conducted on multiple standard HSI datasets have shown that DPL-MSF has significant advantages in scenarios with scarce samples and cross-domain settings. The code will be available at https://github.com/AIYAU/DPL-MSF**

***Index Terms*—Dual-prototype learning, few-shot learning, hyperspectral image (HSI) classification, multisemantic fusion, prompt learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) offer spatial information beyond traditional RGB images and reveal detailed spectral information through continuous spectral bands [1]. The combination of spectral and spatial data makes HSIs a key tool in various fields, such as precision agriculture, geological exploration, environmental monitoring, and military reconnaissance [2], [3]. The high resolution of HSIs can uncover the specific absorption and reflection characteristics of materials, providing a powerful analytical tool for these fields [4]. However, their high-dimensional nature also presents challenges in data processing and analysis [5]. In earlier studies, researchers primarily relied on traditional machine learning algorithms, such as random forests (RFs) [6], k-nearest neighbors (KNNs) [7], and support vector machines (SVMs) [8]. These methods typically require researchers to manually extract features, a process that is not only time-consuming but may also introduce subjective bias, affecting the accuracy and consistency of classification [9], [10]. Furthermore, these traditional methods often focus on spectral information and fail to fully utilize spatial data, leading to an incomplete understanding of image content [11].

To overcome these limitations, deep learning methods have been proposed and widely applied in this field [12]. By automatically learning features from raw data, these methods reduce the need for manual intervention and enhance the depth and accuracy of image understanding [12]. Initially, a deep learning framework that combines spectral and spatial information was proposed by Chen et al. [13], which not only overcame the disadvantages of traditional algorithms but also opened up new possibilities for subsequent research. Inspired by this, several researchers have further explored how to use deep learning techniques for HSI classification. For instance, Ma et al. [14] proposed a feature learning algorithm named contextual deep learning, which explicitly learns spectral and spatial features through deep learning architectures and employs supervised fine-tuning strategies to enhance the performance of feature extractors. In contrast, Li et al. [15] designed a lightweight 3-D convolutional neural network (3D-CNN) that can directly extract deep spectral–spatial features from HSI data without the need for preprocessing or post-processing. This method also achieves the fusion of spectral and spatial information and high-precision classification of HSIs from the perspective of network design. Building on this foundation, Zhong et al. [16] proposed a spectral–spatial residual network (SSRN), which utilizes residual blocks to connect 3-D convolutional layers through identity mapping to enhance the backpropagation of gradients and employs normalization (Norm) to regulate the learning process, thereby improving the classification performance of the trained model. Furthermore, Paoletti et al. [17], addressing the high dimensionality and

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redundancy of HSI data, gradually increased the dimensionality of feature maps through pyramid bottleneck residual blocks, effectively enhancing the network's generalization ability for spectral–spatial features.

Despite the excellent performance of the aforementioned methods, they often rely on a large number of labeled samples. However, the high-dimensional nature of HSIs makes the data annotation process complex and time-consuming [18]. Therefore, few-shot learning methods have been introduced into the field of HSI classification to address this challenge [19]. Researchers mainly tackle the problem of sample scarcity from two perspectives: data augmentation [20] and metalearning [21]. For instance, Haut et al. [22] generated multilevel occluded training images by randomly occluding pixels in different regions of HSI data. Meanwhile, Acción et al. [23] proposed a new data augmentation framework, dual-window superpixels (DWSs), which primarily achieves data augmentation through regional segmentation and independent transformations. These methods effectively address the issue of sample scarcity through data augmentation strategies. On the other hand, Zhang et al. [24] utilized metalearning approaches by introducing prototype vectors as latent variables and employing distribution estimations to obtain their posterior distributions, thereby achieving hierarchical probabilistic reasoning from a Bayesian perspective.

The aforementioned research effectively addresses the issue of sample scarcity from two angles. However, it primarily relies on data from a specific domain and fails to fully leverage shared features from other domains to assist learning, which limits the model's potential for cross-domain knowledge transfer and application [25]. To overcome this limitation, researchers have mainly explored how to use source domain information to improve the learning efficiency and effectiveness in the target domain from the perspectives of domain fine-tuning [26] and domain adaptation [27]. Consequently, Gao et al. [28], based on relational networks, employed domain fine-tuning techniques to fully utilize information from the source domain, achieving effective transfer of cross-domain knowledge. Zhang et al. [29] also used domain fine-tuning methods, finely adjusting the features learned by the model in the source domain to better adapt to the data distribution of the target domain. Peng et al. [30], on the other hand, approached from the perspective of domain adaptation, extracting local and global features and combining them with domain discriminators to effectively reduce domain shift, further enhancing the model's generalization ability on target domain data.

Cross-domain few-shot methods not only alleviate the challenge of sample scarcity but also leverage data information from other domains, achieving cross-domain knowledge transfer and application [31]. However, the data sources of these methods are limited to unimodal information, primarily relying on visual data, and failing to integrate the complementary advantages of multimodal data [32]. In the study of general images, the advent of contrastive language image pre-training (CLIP) has expanded multimodal learning by integrating image and text information [33]. For instance, Pahde et al. [34], based on the prototype network framework, trained a generative model to map text data into the visual feature space to obtain more reliable prototypes. Lin et al. [35], on the other hand, incorporated text information into the image support set on the basis of CLIP, transforming it into an $n + 1$ sample problem for model training. In the field of HSI classification, Li et al. [36] effectively assisted in the generation of image prototypes by utilizing text information in conjunction with contrastive learning methods, and improved classification accuracy by incorporating data augmentation techniques. Another study by Li et al. [37] introduced a collaborative metric strategy to fully exploit text semantics in the decision-making stage.

Existing methods have all incorporated textual information to assist in image prototype generation, but they still exhibit deficiencies in utilizing semantic information. They typically rely solely on class names or brief labels as textual prompts, overlooking the inherent high similarity among label texts. This results in insufficient discriminability of text prototypes and, consequently, limits classification performance in few-shot scenarios. To address this issue, this article proposes a dual-prototype learning with multisemantic fusion (DPL-MSF) approach. Building upon the prototype network [38], our method introduces learnable text embeddings through a prompt learning [39] strategy, enabling dynamic adjustment of textual input and thereby significantly improving the accuracy of text prototypes. Furthermore, we employ a cross-modal alignment strategy to optimize the image prototype generation process via a text-guided semantic enhancement mechanism. In addition, by incorporating soft label (SL) information and a carefully designed feature fusion module (FFM), we effectively integrate image prototypes with SL information, achieving SL-guided semantic refinement and ultimately generating more discriminative class-level prototypes. This multilevel semantic guidance mechanism fully integrates image modality information and label information, not only enriching the semantic representation of class-level prototypes but also achieving significant improvements in classification performance.

The contributions of this article are as follows.

1) To achieve more precise category discrimination, a dual-prototype learning framework based on multisemantic fusion is proposed. This framework effectively addresses the insufficient category discrimination capability of traditional methods in few-shot scenarios through a hierarchical semantic fusion mechanism.
2) A cross-modal alignment method based on learnable text embeddings is proposed, which dynamically adjusts textual representations while maintaining cross-modal feature consistency, significantly improving the model's semantic perception capability in cross-domain scenarios.
3) Soft-label-based semantic correction is proposed. It introduces continuous probability distribution modeling to overcome the discrete representation limitation of traditional one-hot labels and, combined with the feature fusion module, effectively resolves feature confusion between similar categories.

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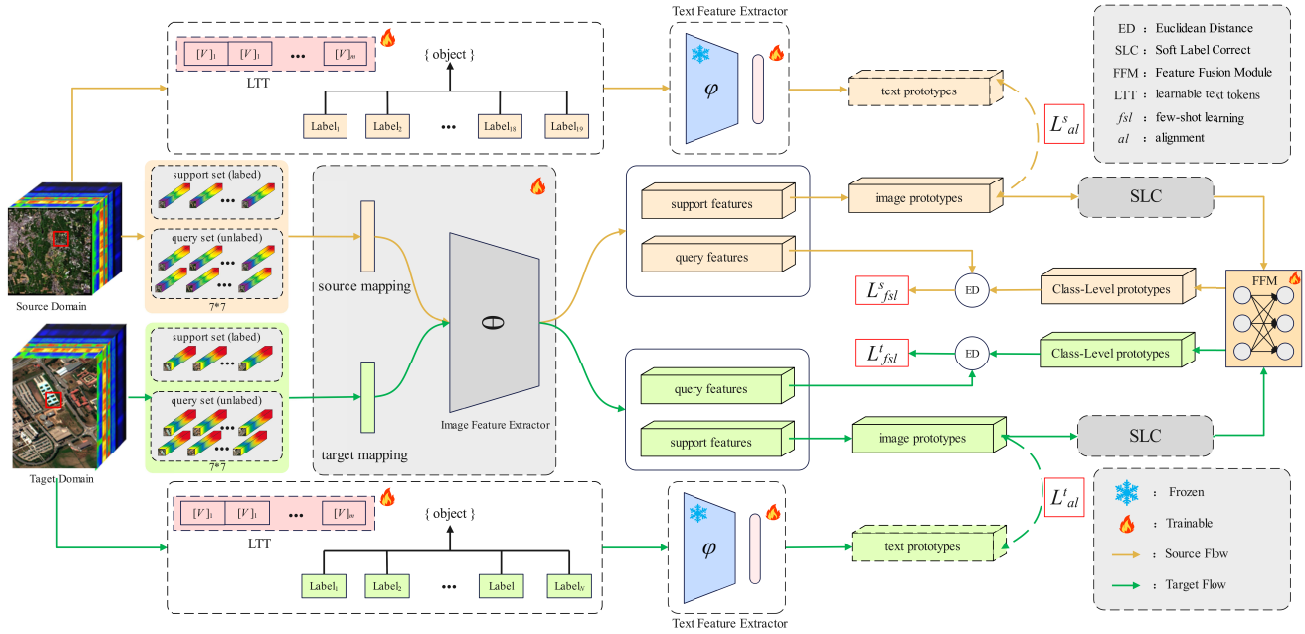


Fig. 1. Training process of the DPL-MSF method. The source and target domains share a feature extractor, and both are optimized simultaneously to effectively transfer knowledge from the source domain to the target domain.

II. Related Works

A. Unimodal Studies

In the field of HSI classification, unimodal methods have achieved significant progress, with technological advancements leading to dual improvements in accuracy and efficiency. For instance, the deep cross-domain few-shot learning (DCFSL) method introduced the conditional adversarial domain adaptation strategy for the first time. By leveraging transferable knowledge from the source domain and minimizing domain shift, it enhances the knowledge representation of target domain images [40]. Building on this, to address the differences between domains, Zhang et al. [41] utilized graph information aggregation to capture nonlocal spatial information, thereby strengthening the model's understanding of nonlocal relationships and enhancing the utilization of source domain knowledge, which in turn strengthens image feature representation. Furthermore, Ye et al. [42] focused on the spectral information of images, using a multiscale spectral–spatial feature extraction module to enhance cross-domain feature representation capabilities with multiscale dilated convolutions. Qin et al. [43] designed multiorder spectral interaction blocks to integrate features into the spectral domain, enhancing the model's discriminative power. His few-shot learning based on feature disentanglement (FDFSL) method also employed orthogonal low-rank feature decoupling technology to further refine image feature representation and optimize the feature space structure, thereby improving classification performance. Liu et al. [52] proposed a multilevel prototype alignment (MLPA) method, which achieves fine-grained classification of cross-domain few-shot HSIs through hierarchical feature alignment and class prototype adversarial training. These strategies emphasize the richness of information in the spectral dimension, making full use of source domain information to achieve feature integration or decoupling in a data-driven manner, thus promoting performance enhancement in HSI classification tasks. However, unimodal information often has limitations and may not fully capture the complexity and diversity of scenes.

B. Multimodal Studies

Multimodal learning methods, by integrating information from various data sources, provide a powerful tool for handling tasks in complex scenarios. Since the emergence of the CLIP model, the field of computer vision has made significant progress in multimodal learning, greatly enhancing the model's ability to understand visual content. For instance, Yan et al. [44] proposed a multimodal transfer feature fusion network. This network uses a dual-branch structure to transfer knowledge from the natural image domain and the HSI domain, and it implements a multitask learning strategy to merge spectral and spatial information. In contrast, Feng et al. [45] integrate HSIs with synthetic aperture radar data, providing an alternative perspective. They achieve the efficient integration of cross-modal information and capture the global dependencies between heterogeneous features through joint intramodal and intermodal contrastive learning, which enhances classification accuracy. However, these methods' reliance on data can lead to increased resource consumption. To address this issue, Li et al. [46] introduced an SGPL method that enhances model performance by incorporating one-hot encoding of semantic information and a prototype generator. Despite improvements in resource consumption, the method's use of one-hot encoding limits its ability to fully capture the data's rich semantic connotations.

III. Methodology

A. Problem Definition

Define the source domain dataset as D_s and the target domain dataset as D_t, where s and t represent the source

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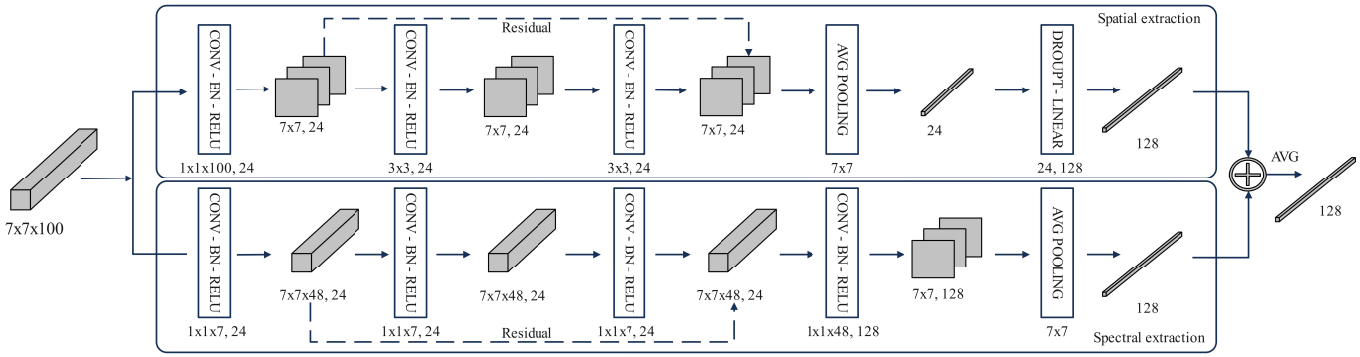


Fig. 2. IFE. A dual-branch architecture extracts features from both spatial and spectral domains, followed by weighted fusion to obtain the final feature representation.

and target domains, respectively. The source domain contains C_s classes, and the target domain contains C_t classes. The experiments follow an episodic training approach, adhering to the N-way-K-shot sampling method. Datasets from all domains are divided into support sets and query sets. Taking the source domain as an example, the support set is defined as $S_s = \{(x_i^s, y_i^s)\}_{i=1}^{N\times K}$, and the query set as $Q_s = \{(x_j^s, y_j^s)\}_{j=1}^{N\times Q}$. Since the experiments involve textual information, a text set is defined as $T_s = \{(t_i^s, y_i^s)\}_{i=1}^{C_s\times 1}$. In addition, the combination of the image feature extractor (IFE) and the mapping layer is defined as $f(\cdot)$, the text feature extractor as $g(\cdot)$, and the FFM as $h(\cdot)$.

B. DPL-MSF Framework

Fig. 1 illustrates the training process of the DPL-MSF method, which mainly includes dataset partitioning, feature extraction, and the execution of a dual-prototype learning strategy. Under the guidance of multiple semantics, dual-prototypes (text prototypes and image prototypes) are continuously optimized to generate class-level prototypes. Specifically, image data is divided into 7×7 pixel patches, which are then fed into a mapping layer to ensure consistency in data dimensions before performing image feature extraction. Through an IFE, initial support set features and query set features are obtained. Support set features undergo a mean aggregation operation to construct preliminary image prototypes. Meanwhile, with the introduction of learnable text tokens (LTTs), text data is input into a text feature extractor to derive refined text prototypes. Image prototypes and text prototypes are aligned through contrastive loss, optimizing the generation process of image prototypes (first-level semantic guidance). At the end of the model architecture, a soft label correct (SLC) strategy is introduced, and under the effect of an FFM, the final class-level prototypes are generated (second-level semantic guidance). During the model optimization process, the semantic information from texts and SLs is deeply integrated, and with the synergistic effect of contrastive loss and few-shot loss, the overall performance of the model is enhanced. Finally, KNN is employed as the classification head, freezing the IFE and mapping layer, and fitting KNN using five samples to complete the classification task. This approach is inspired by Pahde et al. [34], who demonstrated the effectiveness of KNN classifiers in low-data scenarios.

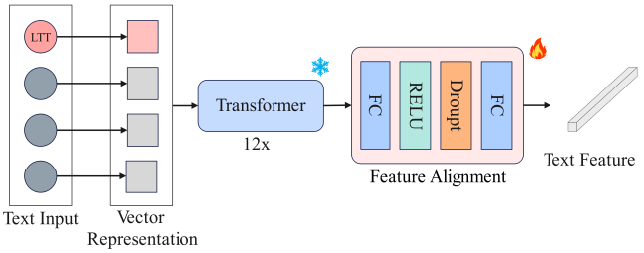


Fig. 3. Text feature extractor. The module consists of a frozen pretrained BERT backbone and a trainable feature alignment layer. LTT means LTT.

C. Feature Extractor

1) Image Feature Extractor: Before the data are input into the IFE, the mapping layer projects the data from the source and target domains into a unified $7 \times 7 \times 100$ tensor space. This step ensures cross-domain dimensional consistency and facilitates the subsequent processing within the IFE. Fig. 2 illustrates the overall architecture of the IFE, which comprises two components for spatial feature extraction and spectral feature extraction, respectively.

In the upper half of the architecture, the initial features undergo dimensionality reduction via a 3-D convolutional layer, during which spectral information is intentionally ignored to focus on extracting spatial features. These features are then further processed by a 2-D residual convolutional network, which enhances their expressive power through a residual learning mechanism [47]. After being processed by an average pooling layer and a linear layer, the spatial feature representations are obtained for subsequent analysis. In the lower half of the architecture, unlike the spatial feature extraction strategy, 3-D convolutions are employed to process spectral features. The core of this part is a 3-D residual block, which effectively captures the dynamic features along the spectral dimension [48]. After deep feature extraction through the 3-D residual network, the features are processed by an average pooling layer to obtain the spectral feature representations for subsequent analysis. Finally, the spatial feature representations are fused with the spectral feature representations with equal weighting, resulting in a unified feature representation that optimizes classification performance.

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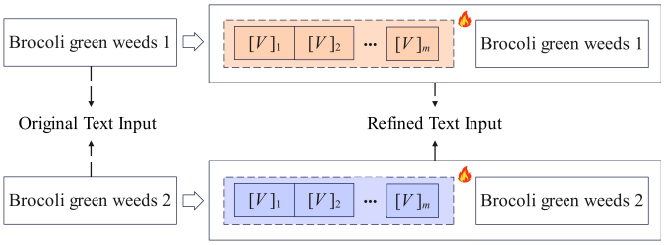


Fig. 4. Refinement of text input.

2) Text Feature Extractor: Fig. 3 illustrates the overall architecture of the text feature extractor. The input text is processed by a parameter-freeze bidirectional encoder representations from transformers (BERTs) base model, which consists of 12 standard transformer encoder layers [49]. This model is designed to leverage its language representation capabilities acquired through pretraining on large-scale corpora. To ensure the output text features' dimensionality matches that of the image features, a feature alignment layer is included. This layer consists of linear transformations that align the dimensions. In addition, the feature alignment layer performs a fine-tuning function, enhancing the model's generalization ability and improving the accuracy of feature representation.

D. Text-Guided Semantic Enhancement

In the DPL-MSF architecture, the first-order semantics correspond to the semantic analysis conducted during the construction phase of text prototypes. Taking the Salinas (SA) dataset as an example, there is a high degree of similarity between certain category names, such as "Broccoli green weeds 1" and "Broccoli green weeds 2." This high similarity means that even after processing by the text feature extractor, the features inevitably retain some degree of similarity. This poses a challenge to the model's precision, requiring it to capture subtle differences between category names. To address this challenge, we introduce LTTs before each category label to optimize the final text input, thereby obtaining more accurate text prototypes. The specific changes to the text input are shown in Fig. 4.

Specifically, we combine the category labels with learnable context tokens to form a new text input, which can be represented as follows:

$$t = [V]_1[V]_2 \dots [V]_M[\text{Label}]. \tag{1}$$

After adjusting the text input, we continue to employ a contrastive learning method, which aims to reduce the distance between positive pairs and increase the distance between negative pairs in the feature space, thereby enhancing the discriminability of image prototypes. Taking the source domain as an example, each category in the text set $T_s = \{(t_i^s, y_i^s)\}_{i=1}^{C_s \times 1}$ has only one text sample. Therefore, the text prototype is directly extracted from the corresponding sample, which can be represented as follows:

$$p_{T_i}^s = g\left(t_i^s\right). \tag{2}$$

Since the experiment employs N-way-K-shot sampling, the image prototype is realized through averaging operations, which can be expressed as follows:

$$p_{I_i}^s = \frac{1}{N} \sum_{x_i^s \in C_k^s} f\left(x_i^s\right) \tag{3}$$

where C_k^s represents the set of support samples of the kth class in the source domain, and $p_{I_i}^s$ represents the image prototype of the source domain. From this, the image prototype set $P_I^s = \{p_{I_1}^s, p_{I_2}^s, \dots, p_{I_N}^s\}$ and the text prototype set $P_T^s = \{p_{T_1}^s, p_{T_2}^s, \dots, p_{T_N}^s\}$ can be constructed. For different prototype representations $p_{T_i}^s$ and $p_{I_i}^s$ of the same class, they are defined as positive sample pairs, while the rest are considered negative samples. Therefore, the construction of the source domain alignment loss can be represented as follows: (4), as shown at the bottom of the page, where $D(,)$ represents the metric based on Euclidean distance, and τ is the temperature coefficient that controls the degree of discrimination. In addition, to consider the alignment problem from multiple perspectives, alignment losses were calculated from both the text perspective and the image perspective. Therefore, the final alignment loss for the source domain can be represented as follows:

$$L_{\text{al}}^s = \frac{1}{2N} \sum_{i=1}^{N} \left[L_{\text{al}}^s\left(x_i^s, t_i^s\right) + L_{\text{al}}^s\left(t_i^s, x_i^s\right)\right] \tag{5}$$

where N represents the batch size. The introduction of the first layer semantics in the target domain is consistent with the operations in the source domain. Hence, the overall alignment loss for the DPL-MSF method can be expressed as follows:

$$L_{\text{al}} = L_{\text{al}}^s + L_{\text{al}}^t. \tag{6}$$

E. SL-Guided Semantic Correction

To address the model's deficiency in capturing subtle semantic differences between categories, a second layer of semantics, namely, SLs, is introduced. This strategy involves converting category labels into one-hot encoding to construct a unique representation for each category in the vector space. When the number of categories in the source and target domains is inconsistent, zero-padding operations are used to ensure the dimensionality of the one-hot encoded vectors is consistent. This is because zero values can be considered as a state of category inactivation, indicating the absence or irrelevance of the category in a specific context. This can be specifically represented as follows:

$$h_{\text{pad}} = h \oplus 0 \tag{7}$$

where h is the original one-hot encoded vector, 0 is the zero vector corresponding to the number of missing categories,

$$L_{\text{al}}^s\left(x_i^s, t_i^s\right) = -\log \frac{\exp\left(-D\left(p_{I_i}^s, p_{T_i}^s\right)/\tau\right)}{\sum_{k=1,k\neq i}^{N} \exp\left(-D\left(p_{I_i}^s, p_{I_k}^s\right)/\tau\right) + \sum_{k=1}^{N} \exp\left(-D\left(p_{T_i}^s, p_{T_k}^s\right)/\tau\right)} \tag{4}$$

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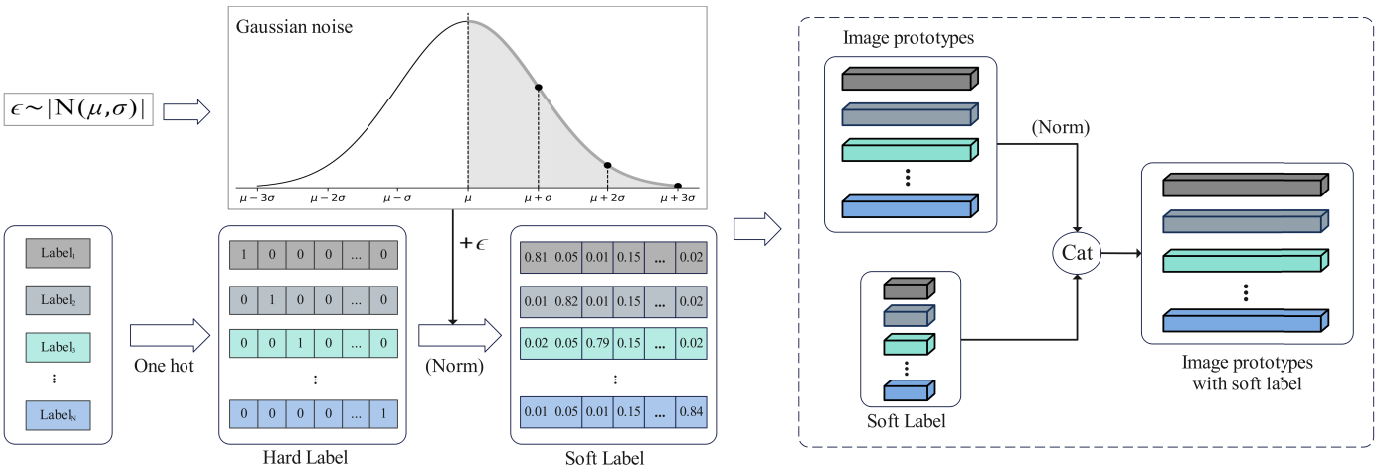


Fig. 5. Soft labels correction Strategy. We introduce positive Gaussian noise to soften hard labels, followed by Norm to generate SLs. The generated SLs are concatenated with image prototypes to implement the SLC strategy. The corrected prototypes are then further fed into the FFM module for optimization.

and h_{pad} is the one-hot encoded vector after the zero-padding operation.

The one-hot encoding of category labels, while providing clear category distinctions mathematically, is inherently absolute and insufficient to capture subtle semantic differences between categories. To overcome this limitation, this study adopts an SL strategy, which incorporates Gaussian noise into the one-hot encoded vectors. The absolute value of the Gaussian noise is used to adjust the one-hot encoding to ensure nonnegativity of the perturbation terms. This operation, while maintaining the validity of the probability distribution, provides the model with a more flexible category representation. It can be specifically represented as follows:

$$s = h_{\text{pad}} + \epsilon, \quad \epsilon \sim |\mathcal{N}(\mu, \sigma)| \tag{8}$$

where ϵ represents the absolute value of the perturbation drawn from the Gaussian distribution, with μ and σ denoting the mean and standard deviation of the Gaussian distribution, respectively. To meet the requirements of the probability distribution, the SLs are normalized to ensure that the sum of their elements is 1. The final setting of the SLs can be represented as follows:

$$s_{\text{norm}} = \frac{s}{\sum s}. \tag{9}$$

Taking the source domain as an example, the fusion of image prototypes and SLs is crucial, with the core being to ensure that both have consistent magnitudes in the vector space. To achieve this, the Norm of image prototypes is realized through the following equation:

$$p_{n_i}^s = \frac{p_{I_i}^s}{\left\| p_{I_i}^s \right\|^2} \tag{10}$$

where $\|p_{I_i}^s\|^2$ is the L_2 norm of the source domain image prototype, and $p_{n_i}^s$ represents the normalized image prototype. The normalized image prototype is concatenated with the SL to complete the preliminary prototype correction, as detailed in Fig. 5.

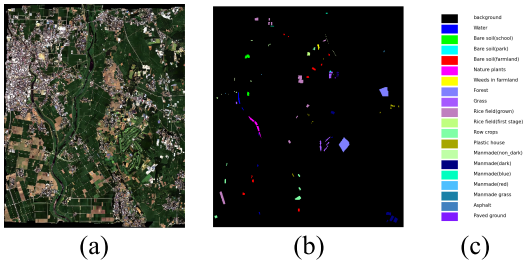


Fig. 6. CH dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) Category of CH.

TABLE I
BASIC INFORMATION OF FIVE DATASETS

Datasets	Source Domain	Target Domain			
	CH	IP	SA	UP	HT
Areas	Chikusei	Indiana	California	Pavia	Houston
Bands	128	102	204	103	144
Spatial size	2517 × 2335	145×145	512 × 217	610×340	349 × 1905
Spectral range	363–1018	400-2500	400-2500	430-860	380–1050

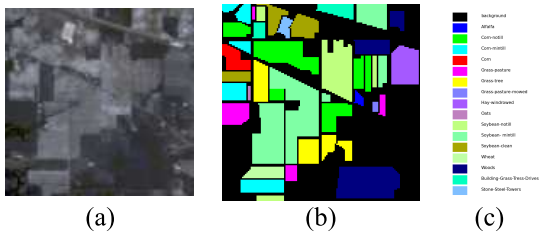


Fig. 7. IP dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) Category of IP.

The feature vector with SL information is further sent into the FFM, as a simple concatenation operation cannot fully integrate the information of these two. The FFM aims to further optimize the feature representation by learning the intrinsic relationships within the data, thereby enhancing the model's ability to recognize subtle differences between

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TABLE II
DETAILED CATEGORY INFORMATION OF SOURCE AND TARGET DOMAIN DATASETS—(CH-IP-SA-UP-HT)

Class	CH		IP		SA		UP		HT	
	Name	Count	Name	Count	Name	Count	Name	Count	Name	Count
1	Water	2345	Alfalfa	46	Brocoli green weeds 1	2009	Asphalt	6631	Healthy grass	1251
2	Bare soil (school)	2859	Corn notill	1428	Brocoli green weeds 2	3726	Meadows	18649	Stressed grass	1254
3	Bare soil (park)	236	Corn mintill	830	Fallow	1976	Gravel	2099	Synthetic grass	697
4	Bare soil (farmland)	48525	Corn	237	Fallow rough plow	1394	Tress	3064	Trees	1244
5	Natural plants	4297	Grass pasture	483	Fallow smooth	2678	Sheets	1345	Soil	1242
6	Weeds in farmland	1108	Grass trees in farmland	730	Stubble	3959	Bare soil	5029	Water	325
7	Forest	20516	Grass pasture mowed	28	Celery	3579	Bitumen	1330	Residential	1268
8	Grass	6515	Hay windrowed	478	Grapes untrained	11271	Bricks	3682	Commercial	1244
9	Rice field (groim)	13369	Oats	20	Soil vinyard develop	6203	Shadow	947	Road	1252
10	Rice field (first stage)	1268	Soybean notill	972	Corn senesced green weeds	3278			Highway	1227
11	Row crops	5961	Soybean mintill	2455	Lettuce romaine 4wk	1068			Railway	1235
12	Plastic house	2193	Soybean clean	593	Lettuce romaine 5wk	1927			Parking Lot 1	1233
13	Manmade (nan-dark)	1220	Wheat	205	Lettuce romaine 6wk	916			Parking Lot 2	469
14	Manmade (dark)	7664	Woods	1265	Lettuce romaine 7wk	1070			Tennis Court	428
15	Manmade (blue)	431	Buildings Grass Trees Drives	386	Vinyard untrained	7268			Running Track	660
16	Manmade (red)	222	Stone Steel Towers	93	Vinyard vertical trellis	1807				
17	Manmade grass	1040								
18	Asphalt	801								
19	Paved ground	145								
Total		77592		10249		54129		42776		15029

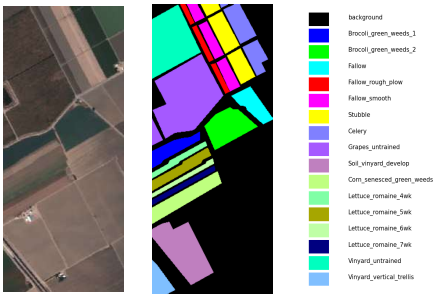


Fig. 8. SA dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) Category of SA.

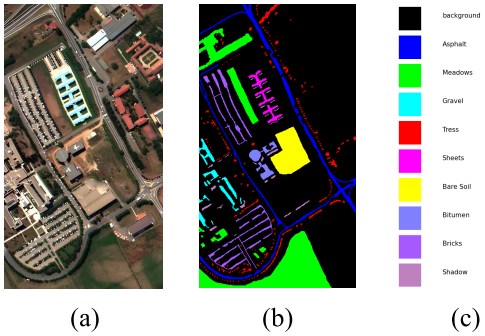


Fig. 9. UP dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) Category of UP.

categories. The FFM is designed as a fully connected layer, which maintains the simplicity of the model while avoiding the risk of overfitting. The rationale for such a design is further demonstrated through experiments in Section IV. The final class-level prototype generated can be represented as follows:

$$p_{c_i}^s = h\left(\left[p_{n_i}^s, s_{\text{norm}}\right]\right). \tag{11}$$

F. Training Loss Function

In the DPL-MSF framework, the class-level prototypes generated are used to measure against the query set $Q_s = \{(x_j^s, y_j^s)\}_{j=1}^{N\times Q}$ to optimize the model. However, it is important to note that the class-level prototypes are scaled prototypes; hence, normalizing the query set is also a necessary operation.

Taking the source domain as an example, the probability of a sample in the query set belonging to the kth class can be represented as follows:

$$p\left(y_j^s = k|x_j^s\right) = \frac{\exp\left(-\text{ED}\left(N\left(f\left(x_j^s\right)\right), p_{c_i}^s\right)\right)}{\sum_{k=1}^{N}\exp\left(-\text{ED}\left(N\left(\left(f\left(x_j^s\right)\right), p_{c_i}^s\right)\right)\right)} \tag{12}$$

where y_j^s represents the sample from the query set, and x_j^s represents the Norm operation. The loss for the source domain is calculated using cross-entropy loss and can be expressed as follows:

$$L_{\text{fsl}}^s = -\sum_{(x,y)\in Q_s}\log p(y = k|x) \tag{13}$$

where L_{fsl}^s denotes the small-sample loss of the source domain, and (x, y) represents the samples in the query set along with their corresponding labels. The operations in the target domain are analogous to those in the source domain, and its small-sample loss can be expressed as L_{fsl}^t Consequently, the overall small-sample loss can be formulated as follows:

$$L_{\text{fsl}} = L_{\text{fsl}}^s + L_{\text{fsl}}^t. \tag{14}$$

When constructing the overall loss function for DPL-MSF, both the few-shot learning loss and alignment loss are taken into account. In the model optimization, the weight of the few-shot loss is set to 1, while the alignment loss is set as an adjustable parameter. This allows for fine-tuning the introduction of the first layer of semantics according to the

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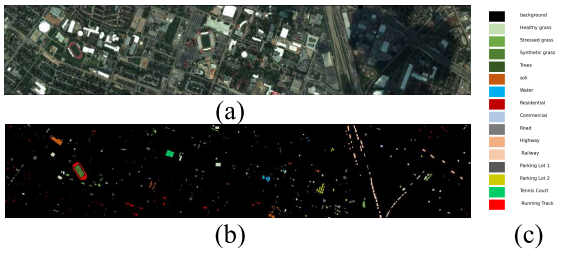


Fig. 10. HT dataset. (a) Pseudo-color image. (b) Ground-truth map. (c) Category of HT.

specific task requirements and data characteristics. Therefore, the overall loss for DPL-MSF can be expressed as follows:

$$L = L_{\text{fsl}} + \lambda \cdot L_{\text{al}}. \tag{15}$$

Algorithm 1 The Training Process of DPL-MSF

Input:

D_s: support set S_s, query set Q_s
D_t: support set S_t, query set Q_t
Text set: T_s and T_t

Output:

Optimize the text input t by Eq. (1)
for episode **do**
Calculate text prototype p_T^s and p_T^t by Eq. (2)
Calculate image prototype p_I^s and p_I^t by Eq. (3)
for $(t_i^s, y_i^s)\} \in T_s$ **do**
Calculate L_{al}^s by Eq. (4) (5)
end for
Calculate L_{al}^t same as L_{al}^s
Calculate L_{al} by Eq. (6)
Generate soft labels by Eq. (7) (8) (9)
for $p_{T_i}^s \in P_T^s$ **do**
Normalize the prototypes by Eq. (10)
Calculate Class-Level prototype $p_{c_i}^s$ by Eq. (11)
end for
Calculate $p_{c_i}^t$ same as $p_{c_i}^s$
for $(x_j^s, y_j^s) \in Q_s$ **do**
Calculate L_{fsl}^s by Eq. (12) (13)
end for
Calculate L_{fsl}^t same as L_{fsl}^s
Calculate L_{fsl} by Eq. (14)
Return total Loss L by Eq. (15)
Optimize the Model by L
end for

To more clearly present the algorithm's process and logic, we have detailed the step division and explanation in Algorithm 1.

IV. Experiments

A. Experimental Datasets

The experimental design addresses a cross-domain issue; hence, the dataset is divided into a source domain dataset and a target domain dataset. We selected Chikusei (CH) [50] as the source domain dataset, which is a widely adopted choice. For the target domain, we chose four public datasets: Indian Pines (IP), SA, University of Pavia (UP), and Houston (HT) to evaluate the effectiveness of our method. All of the above datasets adhere to the few-shot experimental setup. Detailed information about the shooting locations and the number of bands for the five datasets can be found in Table I. Specific sample information for the datasets is presented in Table II In addition, Figs. 6–11 display the pseudo-color images, ground-truth maps, and category information for the five datasets.

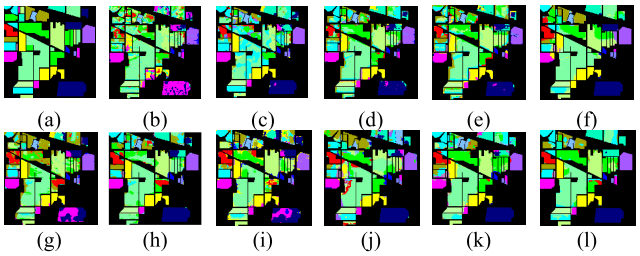


Fig. 11. Comparison of classification effects on the IP dataset. (a) Ground-truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) Gia-CFSL. (f) HFSL. (g) ADAFSL. (h) RPCL. (i) MLRL. (j) FDFSL. (k) APMN. (l) Ours.

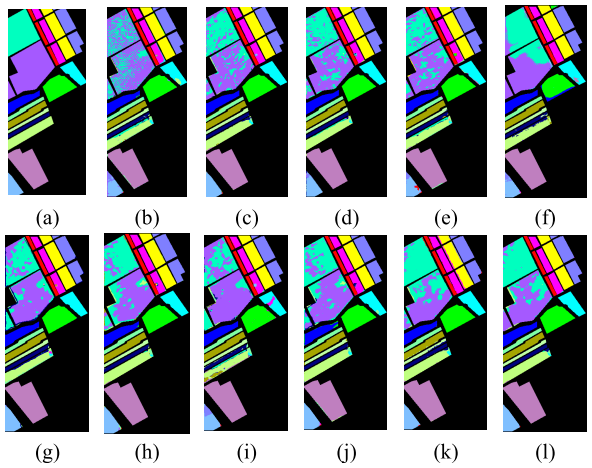


Fig. 12. Comparison of classification effects on the SA dataset. (a) Ground-truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) Gia-CFSL. (f) HFSL. (g) ADAFSL. (h) RPCL. (i) MLRL. (j) FDFSL. (k) APMN. (l) Ours.

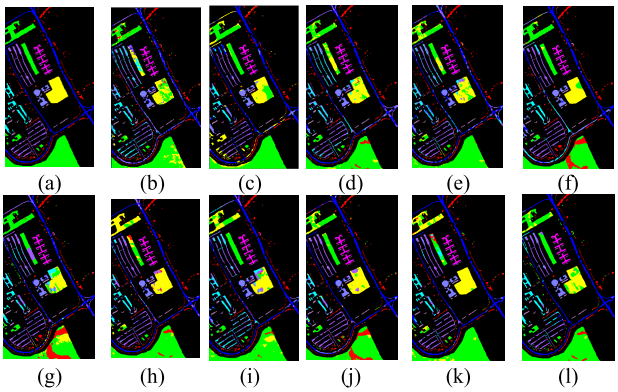


Fig. 13. Comparison of classification effects on the UP dataset. (a) Ground-truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) Gia-CFSL. (f) HFSL. (g) ADAFSL. (h) RPCL. (i) MLRL. (j) FDFSL. (k) APMN. (l) Ours.

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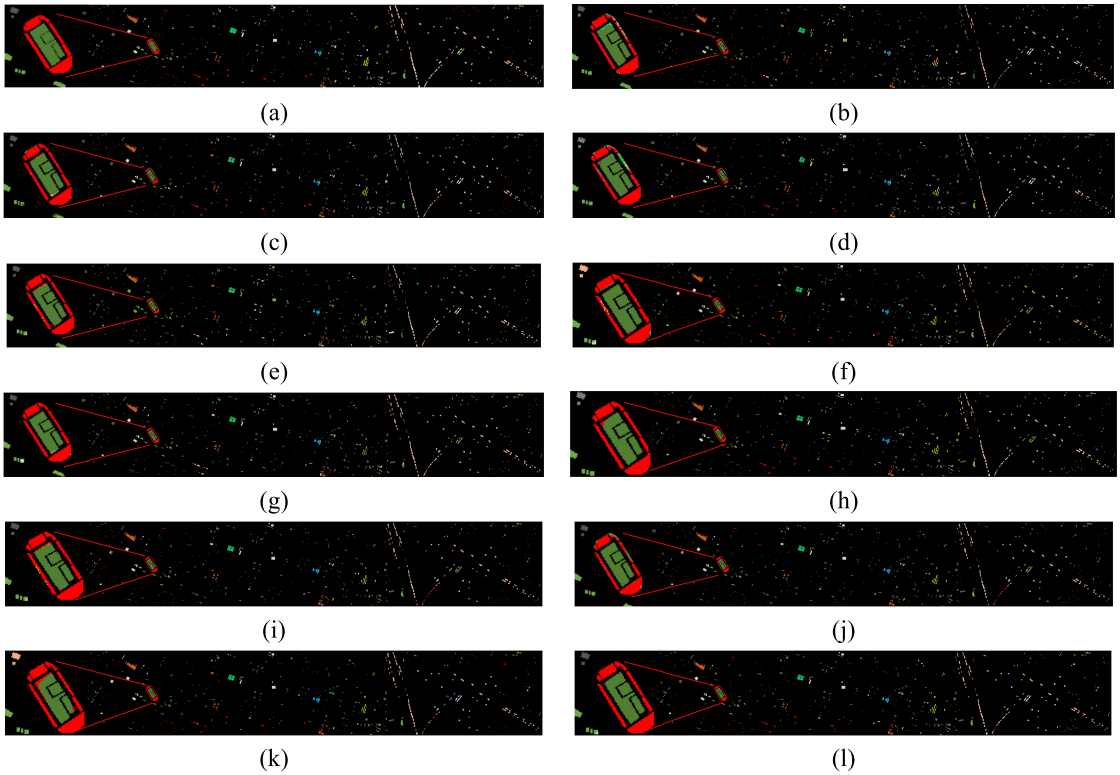


Fig. 14. Comparison of classification effects on the HT dataset. (a) Ground-truth. (b) 3DCNN. (c) SSRN. (d) DCFSL. (e) Gia-CFSL. (f) HFSL. (g) ADAFSL. (h) RPCL. (i) MLRL. (j) FDFSL. (k) APMN. (l) Ours.

TABLE III
PARAMETER TUNING OF THE LTTS

Datasets	Learnable Text Tokens				
	1	2	3	4	5
IP	81.00	**81.73**	81.07	80.83	81.42
SA	93.01	**93.33**	92.94	92.91	93.13
UP	83.48	83.62	**84.60**	83.41	83.64
HT	79.20	**79.80**	79.19	79.00	79.08

TABLE IV
PARAMETER TUNING OF SL STANDARD DEVIATION

Datasets	σ in Soft Label				
	0.000	0.001	0.005	0.010	0.015
IP	81.02	81.04	**81.73**	81.43	81.30
SA	93.21	**93.33**	93.15	93.02	92.99
UP	83.95	83.94	**84.60**	84.28	83.95
HT	79.26	79.19	79.24	**79.80**	79.41

TABLE V
PARAMETER TUNING OF THE FFM'S DROPOUT

Datasets	Dropout of the FFM				
	0.2	0.3	0.4	0.5	0.6
IP	80.95	81.16	81.21	**81.73**	81.37
SA	92.99	93.00	93.25	**93.33**	92.97
UP	84.09	83.88	83.79	**84.60**	83.23
HT	79.58	79.32	79.37	**79.80**	78.94

TABLE VI
PARAMETER TUNING OF λ IN L_{al}

Datasets	λ in L_{al}					
	0.5	1.0	1.5	2.0	2.5	3.0
IP	81.01	**81.73**	81.37	81.05	81.00	81.02
SA	92.71	93.15	**93.33**	92.92	92.84	92.55
UP	83.69	**84.60**	83.61	84.39	84.21	84.24
HT	79.29	79.43	79.25	79.07	**79.80**	79.64

B. Experimental Setups

The hardware configuration for this experiment is as follows: The GPU uses NVIDIA RTX4090 (24 GB), and the computer memory is 128 GB. The experimental code is built under the PyTorch 1.13.1 framework, with CUDA version 11.7 and Python version 3.8.

In the DPL-MSF method, both the source and target domain datasets are divided into 7×7 patch sizes as the initial input for the model. In the few-shot learning (FSL) task, a total of 5000 episodes are trained. Each episode conducts N-way-K-shot training, where N is taken as the number of classes in the target domain (e.g., IP and SA set 16, UP set 9, and HT set 15), and K is set to one, meaning that one labeled sample is used as the support set. In addition, 19 unlabeled samples are used as the query set, and these settings are all followed. The experiment uses Adam as the optimizer to optimize the model and sets the initial learning rate to 0.001. At the end of the experiment, three indicators are used to evaluate the effectiveness of the model, which are overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). To objectively reflect

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TABLE VII
COMPARISON RESULTS (%) BASED ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	3DCNN	SSRN	DCFSL	Gia-CFSL	HFSL	ADAFSL	RPCL	MLRL	FDFSL	APMN	Ours
1	85.04	99.01	97.80	93.41	**100**	97.80	99.76	98.29	95.37	**100**	99.51
2	31.39	42.91	45.14	47.98	66.34	52.18	64.04	48.73	53.89	67.93	**70.85**
3	30.22	58.84	46.80	49.35	56.36	60.16	62.86	62.33	64.08	61.18	**73.08**
4	37.82	81.49	75.60	76.51	**97.63**	86.77	88.97	88.62	83.02	96.03	96.68
5	62.50	78.63	67.74	66.65	65.48	78.70	80.06	78.56	74.12	73.83	**80.21**
6	62.91	95.75	86.50	84.14	86.48	86.03	88.43	88.99	87.24	90.00	**96.79**
7	92.23	99.55	99.57	99.57	**100**	99.57	99.57	99.57	99.57	**100**	**100**
8	59.86	76.90	76.03	83.76	**100**	85.31	90.97	82.03	86.77	95.26	96.34
9	95.24	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	99.33	**100**
10	50.78	60.51	57.08	58.23	62.31	69.71	71.96	67.81	63.02	72.54	**77.53**
11	51.70	51.14	61.45	61.32	**78.41**	58.66	64.76	61.37	63.83	71.66	76.07
12	27.27	62.59	48.69	45.83	70.32	81.04	59.66	52.82	54.69	68.40	**83.44**
13	91.34	99.29	95.35	96.45	98.75	97.65	97.75	98.90	98.95	98.30	**99.55**
14	59.24	87.98	89.48	87.83	**90.95**	84.10	90.06	88.44	87.75	89.98	90.52
15	46.15	75.21	67.72	70.21	82.68	79.50	85.80	74.91	76.48	**87.27**	85.54
16	81.08	97.70	96.82	96.0	92.61	98.30	97.39	**99.77**	97.95	91.36	98.30
OA	49.02 ±4.18	65.72 ±3.77	64.89 ±3.50	65.44 ±2.23	76.75 ±1.80	68.61 ±3.54	74.20 ±2.57	69.38 ±3.87	70.20 ±2.66	77.01 ±1.97	**81.73 ±1.65**
AA	60.30 ±2.30	79.22 ±1.61	75.74 ±2.44	76.08 ±1.48	84.27 ±1.23	80.34 ±1.96	83.88 ±1.28	80.70 ±1.22	80.42 ±1.95	85.19 ±1.09	**89.03 ±1.39**
Kappa	42.63 ±4.28	61.72 ±3.90	60.36 ±3.96	61.02 ±2.53	73.61 ±1.90	64.76 ±3.78	70.92 ±2.83	65.70 ±4.11	66.48 ±2.94	74.08 ±2.14	**79.32 ±1.83**

the performance of the algorithm, all experiments are evaluated based on the average results of ten random seeds.

C. Parameter Tuning

1) LTT Tuning: Text-guided semantic enhancement hinges on the design of text prototypes. Given the high similarity between category names, the LTTs are particularly crucial. To obtain the most precise text prototypes and extract richer semantic information, detailed experiments on LTTs were conducted, as shown in Table III. Studies on the IP, SA, and HT datasets revealed that using two LTTs achieves optimal performance. However, for the UP dataset, three LTTs were found to be the best choice for performance. This further confirms the importance of the learnable tokens in text prototype design and also reveals the challenge of the UP dataset in distinguishing category names, which requires us to allocate more training tokens to improve discrimination for this dataset.

2) Standard Deviation Tuning in SL: Semantic refinement guided by SLs hinges on the design of the SLs themselves. The implementation of SLs is based on one-hot encoding, with the incorporation of Gaussian noise on top of this foundation. We set the mean of the noise, denoted as μ, to 0, but the choice of different standard deviations, denoted as σ significantly affects the quality of the SLs, thereby indirectly impacting the accuracy of semantics. In Table IV, we set the range of σ from 0.000 (hard label) to 0.015 and conducted experiments on four datasets. For the IP and UP datasets, we chose to be 0.005; for the SA dataset, we selected σ to be 0.001; and for the HT dataset, we opted for σ to be 0.010. These results further indicate that different datasets have varying requirements for

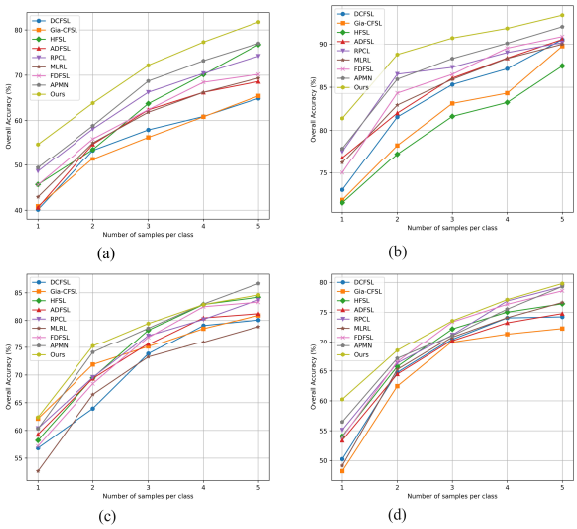


Fig. 15. Comparison experiment in 1–5 shots. (a) 1–5 shots in IP. (b) 1–5 shots in SA. (c) 1–5 shots in UP. (d) 1–5 shots in HT.

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TABLE VIII
COMPARISON RESULTS (%) BASED ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	3DCNN	SSRN	DCFSL	Gia-CFSL	HFSL	ADAFSL	RPCL	MLRL	FDFSL	APMN	Ours
1	93.08	98.53	99.45	98.09	**99.98**	99.63	99.62	99.97	98.86	98.39	99.84
2	95.99	99.67	98.82	99.40	90.84	99.96	99.90	98.75	99.27	99.95	**100**
3	89.80	96.87	97.31	95.46	98.00	97.75	91.04	88.73	91.21	**99.03**	95.89
4	98.15	99.31	99.26	98.52	98.99	**99.83**	98.98	99.21	98.91	96.91	99.24
5	92.34	95.79	91.52	86.55	91.01	96.20	91.67	91.12	94.13	93.99	**97.01**
6	98.80	99.66	99.84	99.25	99.30	99.84	97.90	99.59	98.97	97.94	**99.98**
7	97.83	99.93	99.69	99.28	98.20	98.75	99.36	99.59	99.08	98.78	**99.96**
8	59.31	66.17	79.48	81.31	78.86	77.51	75.00	80.63	79.37	81.31	**83.60**
9	96.93	99.79	99.32	97.76	99.40	99.52	99.69	99.69	98.59	99.81	**99.86**
10	78.10	90.73	86.78	83.73	90.97	85.93	88.41	85.83	85.50	**92.15**	91.41
11	89.85	97.34	97.01	93.56	98.93	**99.72**	98.57	97.66	96.39	99.22	99.60
12	99.34	99.88	99.17	99.19	97.92	99.83	98.60	98.73	99.39	96.72	**99.98**
13	95.17	99.01	97.85	94.27	99.20	**99.49**	98.85	98.49	97.67	97.60	99.09
14	91.24	98.13	98.43	98.46	94.18	98.56	97.43	98.15	**99.52**	93.35	97.65
15	59.06	81.47	75.73	74.51	58.55	75.05	81.44	72.62	81.23	80.91	**83.85**
16	83.21	94.43	92.75	90.32	89.03	97.21	91.77	92.33	91.54	**98.60**	95.58
OA	81.74 ±2.31	89.13 ±2.59	90.51 ±0.99	89.75 ±0.80	87.47 ±1.16	90.52 ±1.90	90.15 ±2.32	89.97 ±1.73	90.85 ±0.90	91.99 ±0.63	**93.33 ±0.50**
AA	88.64 ±1.57	94.79 ±0.55	94.52 ±0.66	93.10 ±1.19	92.71 ±0.43	95.30 ±0.99	94.26 ±1.16	93.82 ±1.54	94.35 ±0.98	95.29 ±0.80	**96.41 ±0.79**
Kappa	79.72 ±2.53	87.95 ±2.80	89.45 ±1.09	88.60 ±0.90	86.04 ±1.33	89.48 ±2.09	89.07 ±2.53	88.83 ±1.92	89.83 ±0.99	91.09 ±0.68	**92.58 ±0.57**

TABLE IX
COMPARISON RESULTS (%) BASED ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	3DCNN	SSRN	DCFSL	Gia-CFSL	HFSL	ADAFSL	RPCL	MLRL	FDFSL	APMN	Ours
1	59.82	**91.84**	78.09	73.25	64.78	80.56	84.89	77.65	86.07	85.21	91.55
2	63.05	**95.13**	81.63	84.95	89.02	81.72	82.67	81.39	83.09	82.91	76.08
3	68.91	55.23	64.72	62.70	75.56	72.09	72.48	61.55	65.16	83.51	**85.58**
4	77.31	78.02	**94.65**	94.05	88.76	87.90	94.82	87.81	89.29	91.99	91.02
5	90.77	98.34	99.38	99.65	**100**	98.45	99.19	99.81	99.40	99.57	99.99
6	63.40	53.56	74.94	73.73	84.72	73.70	77.80	69.49	75.11	**93.93**	87.90
7	87.64	60.07	78.44	81.94	95.38	87.05	81.14	82.48	88.66	**98.67**	98.56
8	57.27	85.34	68.25	72.34	82.70	76.84	83.59	70.15	85.31	85.58	**90.60**
9	95.57	98.08	98.62	96.18	94.61	99.11	98.72	96.88	94.62	91.26	**98.96**
OA	65.74 ±1.77	76.26 ±5.78	80.08 ±3.92	80.91 ±1.29	84.20 ±4.15	81.22 ±4.52	83.72 ±2.77	78.88 ±3.65	83.31 ±1.89	**86.67 ±3.47**	84.60 ±3.60
AA	73.72 ±1.01	79.51 ±3.21	82.08 ±1.91	82.09 ±2.04	86.17 ±2.40	84.16 ±2.57	86.14 ±2.72	80.80 ±2.11	85.19 ±2.23	90.29 ±1.90	**91.94 ±1.37**
Kappa	57.37 ±1.97	70.56 ±6.69	74.46 ±4.45	75.33 ±1.53	79.53 ±2.40	75.82 ±5.15	79.02 ±3.23	72.80 ±4.19	78.84 ±2.16	**82.55 ±4.18**	80.50 ±4.15

the introduction of SLs, which in turn affects the effectiveness of model training and the final classification accuracy.

3) Dropout Tuning in FFM: The design of the FFM is crucial for achieving effective integration of SLs with image prototypes. Therefore, experiments on the Dropout parameter of FFM within a range of 0.2–0.6 were conducted in Table V to optimize the performance of DPL-MSF. It can be observed in the table that for the four datasets (IP, SA, UP, and HT), the model achieves optimal performance when Dropout is set to 0.5.

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TABLE X
COMPARISON RESULTS (%) BASED ON THE HT DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	3DCNN	SSRN	DCFSL	Gia-CFSL	HFSL	ADAFSL	RPCL	MLRL	FDFSL	APMN	Ours
1	89.24	81.94	**90.20**	89.66	74.17	89.56	90.37	88.68	87.38	84.14	87.16
2	77.15	80.72	74.45	74.43	56.32	87.56	86.27	83.75	**88.06**	85.83	86.76
3	95.06	99.68	94.65	93.48	96.50	97.33	96.39	96.40	97.90	98.90	**99.73**
4	88.15	89.81	91.38	89.18	89.80	89.96	92.22	92.12	92.07	90.86	**92.70**
5	94.68	95.78	96.90	94.69	87.92	95.73	97.04	94.77	**99.42**	99.06	97.75
6	72.12	83.57	84.34	79.61	**88.81**	78.34	85.28	82.44	85.06	86.56	88.38
7	54.53	66.79	59.29	58.38	85.67	60.84	**80.13**	73.50	69.70	75.34	79.14
8	42.38	22.36	42.59	37.74	42.92	42.24	43.54	42.18	**48.42**	44.23	39.91
9	59.98	61.50	59.95	55.93	68.04	56.12	**74.39**	68.91	73.17	67.44	73.17
10	53.29	49.14	66.79	60.37	71.95	**68.54**	65.34	62.63	64.97	71.91	68.41
11	61.13	77.97	56.02	59.46	**80.62**	65.55	69.11	60.33	64.52	69.89	75.90
12	44.77	40.70	**75.42**	71.84	71.50	71.87	64.09	65.18	62.14	70.64	61.55
13	31.56	75.09	86.73	88.64	90.52	44.94	**95.45**	91.25	93.00	93.79	91.25
14	86.36	99.80	88.60	87.42	99.62	94.52	88.98	89.67	98.16	99.86	**99.93**
15	96.56	99.77	94.65	91.41	90.66	94.99	97.21	97.05	99.39	94.81	**100**
OA	68.58 ±2.58	70.46 ±1.24	74.22 ±2.92	72.26 ±3.60	76.41 ±2.20	74.77 ±2.85	79.29 ±2.15	76.63 ±2.28	78.57 ±1.98	79.30 ±2.80	**79.80 ±2.22**
AA	69.80 ±2.61	74.97 ±0.99	77.47 ±2.62	75.48 ±2.82	79.67 ±1.64	75.87 ±3.01	81.72 ±1.76	79.30 ±2.10	81.56 ±1.55	82.22 ±2.29	**82.78 ±1.80**
Kappa	65.92 ±2.76	68.08 ±1.33	72.16 ±3.16	70.05 ±3.89	74.52 ±2.38	72.72 ±3.08	77.64 ±2.32	74.75 ±2.47	76.86 ±2.12	77.64 ±3.02	**78.20 ±2.39**

TABLE XI
ABLATION STUDY ON LAYER NUMBERS IN FFM

Datasets	Layers (FFM)			
	1	2	3	4
IP	**81.73**	81.56	80.93	80.87
SA	**93.33**	93.20	92.88	92.43
UP	**84.60**	84.12	83.52	83.10
HT	**79.80**	79.25	78.26	78.19

TABLE XII
ABLATION STUDY IN DIFFERENT MODULES

IFE	Image Prototype		Text Prototype		IP	SA	UP	HT
	Norm	SL	TPA	LTT				
√					75.43	91.20	82.86	78.20
√	√				78.52	91.75	83.10	78.53
√		√			78.43	92.10	82.93	78.69
√			√		79.68	92.41	83.64	79.40
√	√	√			80.56	92.55	83.15	79.15
√			√	√	80.60	93.10	83.92	79.64
√	√	√	√	√	**81.73**	**93.33**	**84.60**	**79.80**

4) Loss Weight Tuning: In the experiments, the total loss is jointly composed of the contrastive loss and the few-shot loss. The L_{fsl} as core loss for the classification task is always assigned a weight ratio of 1. Regarding the L_{al}, to more effectively integrate the first layer of semantic information, we not only considered the LTTs but also paid special attention to the configuration of the loss weights. Therefore, in Table VI, we conducted an analysis experiment on the weight ratio of the text from 0.5 to 3.0 to optimize its contribution to the model performance. For the IP and UP datasets, the model achieved the best performance when the contrastive loss weight ratio was set to 1. For the SA and HT datasets, the optimal performance was reached when the contrastive loss was set to 1.5 and 2.5, respectively.

D. Comparison With Other Methods

To comprehensively evaluate the performance of DPL-MSF under few-sample conditions, we compared it with ten carefully selected methods that cover a wide technical spectrum—from fundamental to cutting-edge approaches. The compared methods include: 3DCNN [15], which establishes a fundamental 3-D convolutional framework for joint spectral–spatial feature learning; SSRN [16], which improves training stability through residual connections; DCFSL [40], an early adversarial domain adaptation approach for cross-domain alignment; graph information aggregation cross-domain few-shot learning (Gia-CFSL) [41], which introduces graph aggregation to capture global contextual relationships; heterogeneous few-shot learning (HFSL) [53], a heterogeneous transfer method leveraging natural image

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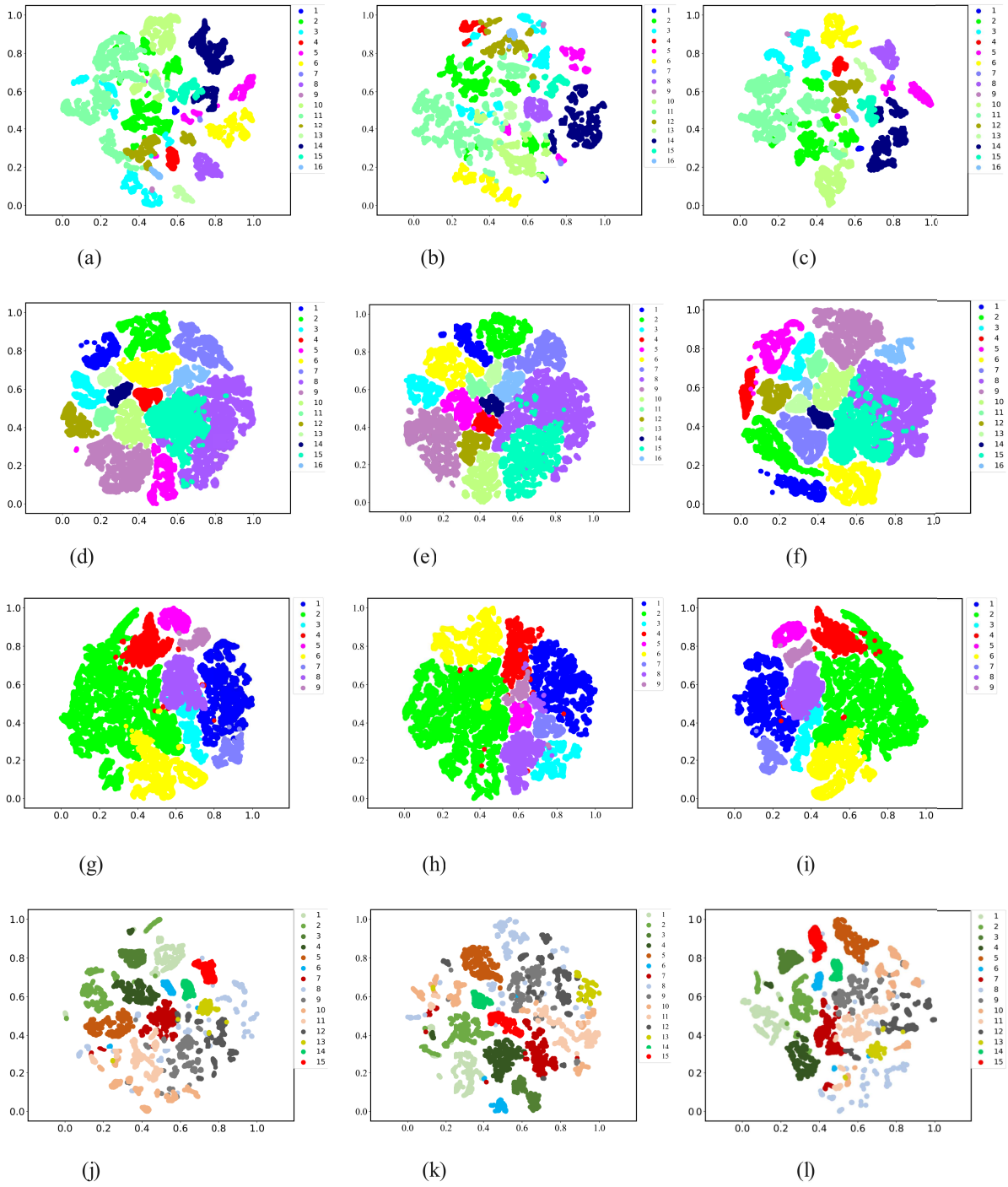


Fig. 16. t-SNE distribution analysis compared with SOTA methods. (a) FDFSL in IP. (b) APMN in IP. (c) Ours in IP. (d) FDFSL in SA. (e) APMN in SA. (f) Ours in SA. (g) FDFSL in UP. (h) APMN in UP. (i) Ours in UP. (j) FDFSL in HT. (k) APMN in HT. (l) Ours in HT.

knowledge; adaptive domain-adversarial few-shot learning (ADAFSL) [42], which employs multiscale dilated convolution for spectral–spatial refinement; refined prototypical contrastive learning (RPCL) [54], a contrastive learning-based method with prototype calibration; multilevel relation learning (MLRL) [51], which incorporates transformer for multilevel relation modeling; FDFSL [43], integrating spectral interactions and feature decoupling; and attention based patch mixing network (APMN) [55], the latest approach using patch mixing and self-attention to enhance sample diversity and discriminability. Together, these methods represent both foundational and state-of-the-art (SOTA) advances in few-shot HSI classification, ensuring a thorough and fair evaluation across major technical directions.

Tables VII–X detail the comparative results across four datasets (IP, SA, UP, and HT) for different methods. On the IP dataset, our method (DPL-MSF) achieves OA, AA, and Kappa scores of 81.73%, 89.03%, and 79.32%. Compared with the

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TABLE XIII
COMPUTATIONAL COMPLEXITY ANALYSIS WITH DIFFERENT METHODS

Dataset	Metrics	Method										
		3DCNN	SSRN	DCFSL	Gia-CFSL	HFSL	ADAFSL	RPCL	MLRL	FDFSL	APMN	Ours
IP	Training time	112.78s	159.64s	2101.45s	4269.32s	6938.56s	2501.45s	435.17s	3633.43s	2453.21s	488.23s	1951.17s
	Testing time	4.26s	13.56s	5.67s	2.56s	57.34s	6.53s	1.44s	1.73s	5.79s	1.23s	3.36s
	#Params	0.10M	0.35M	4.28M	0.86M	6.27M	3.21M	4.27M	6.08M	5.28M	0.36M	0.33M
SA	Training time	152.36s	219.34s	2760.51s	4252.35s	7312.02s	3120.12s	581.66s	3636.96s	3136.15	573.14s	2340.22s
	Testing time	15.27s	52.17s	23.54s	10.25s	163.15s	21.25s	8.94s	8.97s	26.78s	7.12s	7.27s
	#Params	0.10M	0.35M	4.28M	0.86M	6.27M	3.21M	4.27M	6.08M	5.28M	0.36M	0.33M
UP	Training time	53.17s	67.52s	1534.98s	4001.86s	1352.46s	2980s	368.12s	2843.12s	2857s	447.50s	2150.43s
	Testing time	6.25s	31.43s	6.84s	5.33s	7.40s	19.23s	1.55s	5.50s	15.43s	4.54s	5.23s
	#Params	0.07M	0.26M	4.27M	0.85M	6.19M	3.21M	4.26M	6.07M	5.26M	0.32M	0.32M
HT	Training time	99.47s	113.56s	1856.23s	3253.28s	1232.26s	2563.4s	333.24s	2438.25s	1832.52s	502.74s	1531.47s
	Testing time	4.12s	10.25s	4.68s	1.98s	5.28s	5.17s	1.55s	3.65s	4.32s	5.27s	1.72s
	#Params	0.07M	0.26M	4.27M	0.85M	6.19M	3.21M	4.26M	6.07M	5.26M	0.32M	0.32M

other ten methods, DPL-MSF improves OA by at least 4.72 percentage points, and boosts AA and Kappa by 3.84 and 5.24 percentage points, respectively. DPL-MSF further attains 93.33% OA, 96.04% AA, and 92.58% Kappa, corresponding to gains of 1.34, 1.12, and 1.49 percentage points. On the UP dataset, DPL-MSF ranks second with 84.60% OA and 80.50% Kappa, while its 91.94% AA improves by 1.65 percentage points. This is attributed to DPL-MSF's superior robustness under class imbalance, which markedly enhances the accuracy of minority classes such as shadow and sheets, thereby elevating the AA metric. Conversely, its slightly weaker global feature modeling reduces discrimination of majority classes like Meadows, causing minor drops in OA and Kappa while preserving overall second-best competitiveness. On the HT dataset, DPL-MSF records 79.80% OA, 82.78% AA, and 78.20% Kappa, corresponding to improvements of 0.5, 0.56, and 0.56 percentage points, respectively. Overall, our method achieves the best performance in most metrics across the four datasets, especially on IP, further corroborating the effectiveness of incorporating multiple semantics.

To more intuitively display the classification performance of different methods, Figs. 12–15 present their comparisons with the ground truth. These plots clearly show that, among all compared approaches, the maps produced by DPL-MSF most closely resemble the ground-truth reference on the IP, SA, and HT datasets. On the UP dataset, DPL-MSF delivers the best results for the small-sample categories.

To comprehensively evaluate the performance of the DPL-MSF method in a low-shot setting, we conducted not only five-shot experiments but also expanded the scope to include 1–4 shot experiments. Across all tested scenarios, DPL-MSF consistently achieved superior results. These comprehensive evaluations not only confirm the potential of DPL-MSF in few-shot learning, but also underscore the significant benefits introduced by leveraging semantic information. Detailed results are presented in Fig. 16.

To more intuitively demonstrate the performance differences in feature extraction among various methods, we plotted the t-distributed stochastic neighbor embedding (t-SNE) diagrams. Through these diagrams, one can visually compare the distribution of feature vectors extracted by different methods in low-dimensional space. In Fig. 16, the feature distribution of the SOTA methods from the past two years and DPL-MSF is primarily displayed. It can be observed that the DPL-MSF method achieves the best distributional performance on the IP, SA, and HT datasets. It also demonstrates a clear advantage on the UP dataset.

E. Ablation Study

1) Layer-Wise Ablation Analysis of FFM: When exploring the efficient integration mechanism of SLs and image prototypes, the architectural design of the FFM is particularly crucial. In addition to the adjustment of the dropout parameter, the choice of the number of layers also has a significant impact. By conducting an ablation study on the number of layers in the FFM, we present the experimental results for four datasets in Table XI. The results indicate that under a single-layer structure, the model achieved optimal performance across all datasets. However, as the number of layers increases, there is a corresponding decrease in performance. This phenomenon can be attributed to the relatively small amount of semantic information contained in SLs compared with image prototypes, with pixel-level information dominating the feature expression. Therefore, a single-layer structure is sufficient to achieve effective integration of SLs and image prototypes. Further increasing the number of layers may lead to model overfitting, thereby disrupting the original information structure of the image and affecting the model's generalization ability.

2) Ablation Study of Modules: The DPL-MSF method employs a dual-prototype learning strategy, and this research conducts an ablation study focusing on the two prototypes and their key operational steps. Within the context of the IFE, the introduction of SLs represents a core step in the image prototype process, which primarily includes two stages: the Norm of the image prototype and the SL correction. It

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should be noted that the introduction of the FFM is specifically designed to facilitate the integration of SLs with image prototypes. Consequently, the FFM and SLs are implemented in conjunction. Hence, we have conducted an ablation experiment exclusively on the SLs. For the text prototype, the focus is on text prototype activation (TPA) and the introduction of LTTs. As illustrated in Table X, each critical operation plays a vital role in boosting the model's performance. Notably, the dual prototypes demonstrate optimal performance when integrated under multisemantic fusion, as implemented in the complete DPL-MSF method. For specific experiments, see Table XII.

F. Computational Complexity Analysis

In the comparative experiments, the DPL-MSF method demonstrated superior performance. To further analyze the differences among the approaches, we conducted a detailed comparison of their computational complexities. As shown in Table XIII, traditional deep models (3D-CNN and SSRN) have fewer parameters but exhibit significantly limited performance in few-shot scenarios. Starting from DCFSL, both parameter count and training time increase, yielding a notable performance boost. In contrast, DPL-MSF maintains leading accuracy while keeping model size, training duration, and inference time at relatively low levels, showcasing a comprehensive advantage. It ensures efficient integration of semantic information and improved model performance while significantly reducing resource consumption, achieving a dual optimization of efficiency and performance.

V. Conclusion

This article proposes a dual-prototype learning framework based on multisemantic fusion for HSI classification. To address the insufficiency of prototype representation in unimodal data, the framework introduces multiple semantics. The first layer of semantics is the text prototype optimized by LTTs, which, through a cross-modal alignment strategy, provides precise guidance for the learning of image prototypes. The second layer of semantics is the refinement of the image prototype by SLs, which, combined with the FFM, deeply integrates both types of information to generate class-level prototypes. With the introduction of multiple semantics, the prototypes become more discriminative and representative. Future work will continue to explore the guiding role of multimodal information in HSI classification, aiming to achieve more efficient and accurate classification models.

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